

PRE-SEED · \$5M TARGET · PLANNED SAFE

Make change the unit of compute

For edge and embedded teams whose battery, heat, bandwidth, latency, or footprint is constrained by repeated state work.

1

workload

Start with one constrained state path.

1

baseline

Measure against the current system.

1

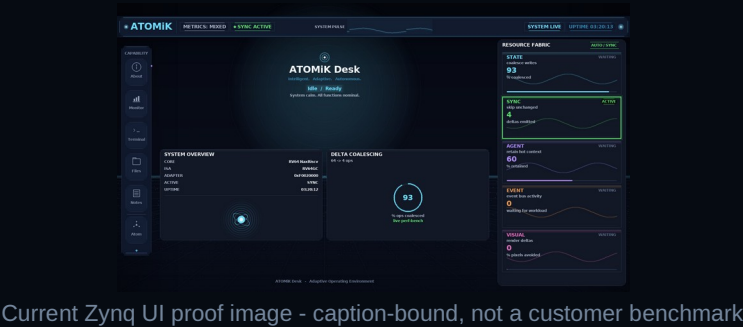
constraint

Battery, heat, latency, bandwidth, footprint, or hardware limit.

\$5M

ask

18-24 month proof round. No tape-out funding.



Current Zynq UI proof image - caption-bound, not a customer benchmark

Public proof image is a hardware-validated UI artifact, not customer workload proof.

PROBLEM

Customers already know the pain

- Batteries die too quickly.
- Hardware is maxed out.
- Systems run hot.
- Devices are too big or too heavy.
- Costs keep increasing.
- More performance is needed.

Before they spend more money

- bigger hardware
- larger batteries or cooling
- more bandwidth or a redesign

ATOMiK checks whether a constrained state path can get more out of the system they already have.

END-GAME VALUE

Customers do not buy delta-state algebra

Bytes

avoided

less state movement against the baseline

Latency

reduced

faster local state paths where fit exists

Ops

coalesced

less repeated work in change-heavy paths

Power

proxied

follow-on measurement when instrumentation exists

Customer ROI is measured after the workload shows reduced state movement while preserving correctness.

MARKET OPPORTUNITY

Large pressure pool, focused wedge, strategic optionality

\$1T+**TAM context**

2026 semiconductor revenue context. Backdrop only; ATOMiK does not address all semiconductor spend.

\$112B-\$169B**SAM context**

Embedded systems market range from 2024 estimate to 2030 forecast; edge AI adds adjacent pressure.

\$10M-\$40M**entry wedge**

early annual revenue path from evaluations, design partners, and licensing; not a ceiling.

Venture logic: proof-backed IP can support licensing and strategic optionality if workload evidence and partner diligence hold.

FOCUS MARKETS

Growth pressure meets constrained state work

Edge AI

\$24.9B 2025 -> \$118.7B 2033

21.7% CAGR

Target examples: NVIDIA Jetson ecosystem, Qualcomm, Ambarella, Hailo, Advantech

Robotics / industrial

\$33.8B 2024 -> \$131.5B 2030

26.5% CAGR

Target examples: Rockwell Automation, Siemens, ABB, FANUC, Universal Robots

Drones / remote autonomy

\$30.0B 2024 -> \$54.6B 2030

10.6% CAGR

Target examples: Skydio, AeroVironment, Anduril, Shield AI, Teledyne FLIR

Smallsat / remote sensing

\$4.0B 2024 -> \$14.0B 2030

22.8% CAGR

Target examples: Planet, Spire, BlackSky, Maxar, Rocket Lab

Sources: Grand View Research segment reports. Target examples are account hypotheses only; no relationship or endorsement implied.

COMPETITIVE LANDSCAPE

Alternatives are real; proof-bound IP is the wedge

More silicon / accelerators

NVIDIA Jetson, Qualcomm, Ambarella, Hailo, Google Coral;
FPGA upgrades from AMD/Xilinx, Intel/Altera, Lattice

Processor and IP incumbents

Arm, Synopsys ARC, Cadence Tensilica, CEVA, Andes/RISC-V ecosystem, Imagination

Software/status quo

compression, caching, dedup, delta sync, CRDT/event sourcing, hand optimization, bigger batteries, more cooling

Defensibility wedge

Lean4-checked formal algebra + FPGA hardware validation + IP/proof registry. Exact formal claims stay tied to audited properties.

PRIMITIVE

Make change the unit of compute

state = reference_state XOR accumulated_delta

LOAD

set reference

ACCUM

add deltas

READ

reconstruct

SWAP

commit boundary

FIT

ATOMiK has a wedge, not a universal claim

- updates dominate reads
- state changes are sparse
- bandwidth or latency is expensive
- power budget or hardware margin is tight
- sync, replay, rollback, or context retention creates overhead

Not the claim

- CPU/GPU/NPU replacement
- universal speedup
- thermal or battery result before measurement

USE CASES

Same waste, different customer budgets

Robotics / industrial

heat, latency, reliability, control-state churn

AI at the edge

context movement, memory pressure, local response

Remote systems

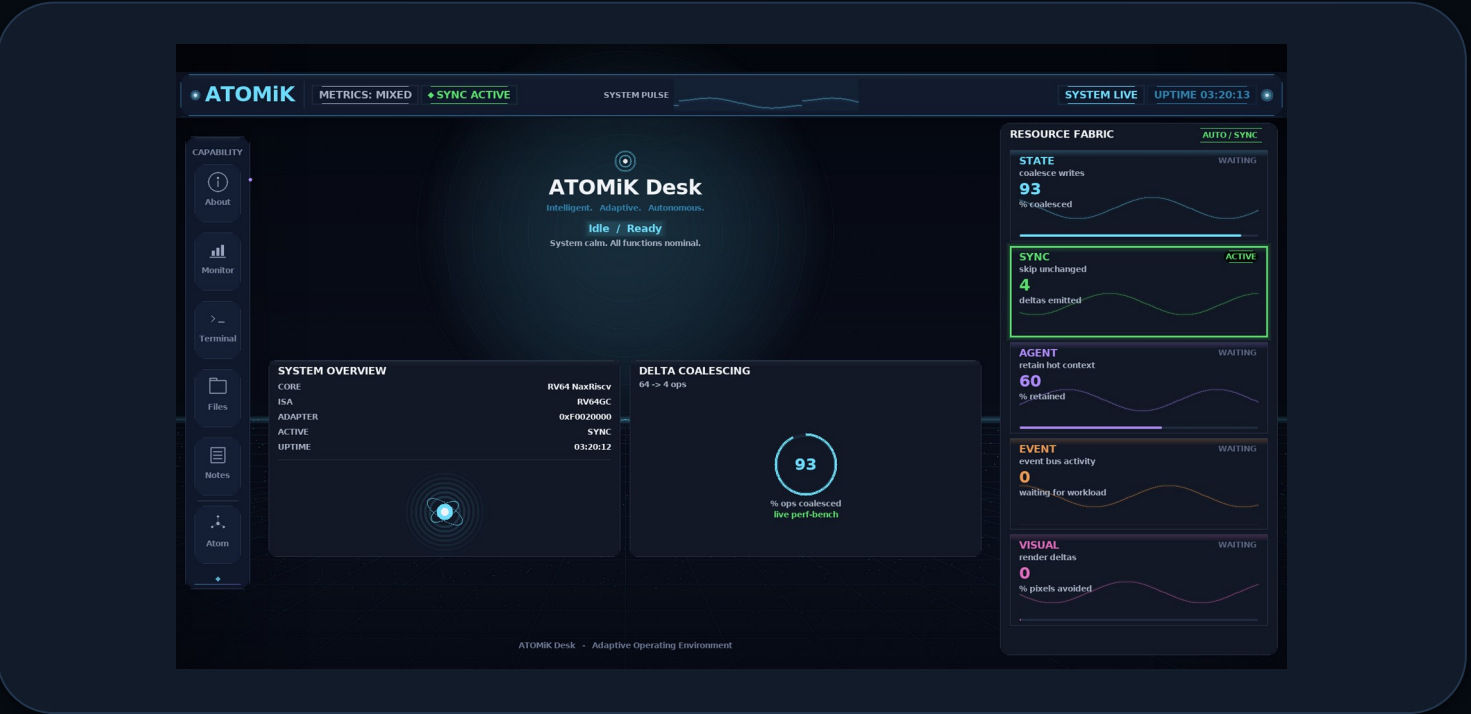
drones, satellites, field runtime, packet budget

Defense-adjacent

constrained hardware, power, rugged reliability

LIVE PROOF

Current Zynq UI artifact with caption



What it shows

current Zynq UI captured from /dev/fb0, driven by real measured on-board data

What to quote

caption-bound hardware-validated UI artifact

What not to quote

on-screen telemetry as customer benchmark; not an interactive demo

PROOF STACK

Real, specific, evidence-bounded

Zynq Desk v0.40-A	HARDWARE_VALIDATED	current UI proof image, captured from /dev/fb0
Parallel-bank throughput	LIVE_MEASURED	AX7020 8-bank accumulator scales 1/2/4/8x, byte-identical to software
Linux userspace to FPGA	HARDWARE_VALIDATED	documented OS-to-bus path
AX7020 board matrix	LIVE_MEASURED	wins and losses with caveats
Formal algebra	FORMAL_PROOF*	Lean4-checked algebra; exact audited properties only
Standalone boot	BUILD_ARTIFACT	not promoted until run proof

*FORMAL_PROOF applies only to specifically audited formal properties; it does not imply workload or customer outcomes.

COMMERCIAL PATH

A customer can understand the first step

1. Show us the problem path

the part causing battery, heat, hardware, latency, size, weight, or cost pressure

2. We evaluate fit

measure whether ATOMiK can reduce state work against the current baseline

3. We show the evidence

where the opportunity is, what was measured, and the potential impact

4. Or call no-fit

if ATOMiK does not help, say so and stop before wasting customer time

Design-partner work and licensing diligence come only after measured proof supports the path.

FINANCIAL MODEL

\$5M funds proof, partners, and licensing diligence

Founder-prepared planning scenario, not a forecast.

\$0-\$150K

0-6 mo

qualified proof reviews / technical evals

\$250K-\$750K

6-12 mo

paid evaluation or design-partner SOWs

\$1M-\$3M

12-24 mo

potential contracted value if proof
converts

\$3M-\$8M

24-36 mo

annualized potential if proof and
economics hold

Use of funds: \$1.5M engineering | \$1.0M customer proof | \$750K IP/legal | \$750K ASIC feasibility | \$1.0M finance/GTM/ops + reserve

RISKS AND GATES

Trust comes from saying what is still unproven

Customer validation

pending; design partners are the wedge

Downstream outcomes

battery/heat/water/ROI need end-to-end measurement

ASIC economics

external feasibility review before tape-out

Incumbents

differentiate by proof, IP, and workload focus

The risks are exactly why this is a proof round, not a tape-out round.

TEAM

Founder-led proof with commercial translation

**Matthew H. Rockwell****Founder & CEO**

Architecture, live proof, product direction, financial model, ask, and technical Q&A.

**Allison Rossi****CMO**

Messaging, positioning, buyer-language, commercialization, and go-to-market support.

Advisor priorities

ASIC/EDA feasibility reviewer | formal-methods reviewer | semiconductor IP/licensing counsel

THE ASK

\$5M target to reach measured workload proof

- \$5M target pre-seed; final SAFE terms CFO/counsel pending
- 18-24 months to measured workload proof, IP packet, ASIC feasibility
- Need: capital, qualified workloads, design-partner introductions

Priority help

not just capital

design-partner introductions plus investor feedback

Money in

\$5M target

Output

proof + IP + feasibility

Upside

licensing path if proof holds